

E-COMMERCE SALES ANALYSIS EXECUTIVE

SUMMARY

Overview

This project presents a comprehensive data analysis of an e-commerce platform's transactional dataset, leveraging Python, MySQL, Pandas, Matplotlib, and Seaborn. The analysis spans multiple dimensions — customer geography, order behavior, product performance, seller revenue, and year-over-year growth to surface actionable business insights from raw operational data.

Dataset & Infrastructure

The analysis integrates 7 relational datasets loaded into a MySQL database (ecommerce):

Table	Description
customers	Customer profiles including city and state
orders	Order records with purchase timestamps
order_items	Line-item details per order
products	Product catalog with category metadata
sellers	Seller registry
payments	Payment records including value and installment data
geolocation	Geographic coordinates by postal code

Data ingestion was automated via a Python pipeline that dynamically inferred SQL schema from CSV column types and bulk-inserted records with NULL-safe handling.

Key Analyses & Findings

1. Customer Geographic Distribution

- Customers span a diverse set of cities and states across the country.
- State-level analysis revealed a highly skewed distribution — a small number of states account for the majority of the customer base, indicating strong regional concentration.

2. Order Volume (2017)

- A point-in-time query confirmed the total number of orders placed in 2017, establishing a baseline for temporal comparisons.

3. Total Sales by Product Category

- Revenue was aggregated across all product categories by joining products, order_items, and payments.
- Results revealed category-level sales concentration, with certain categories significantly outperforming others in total payment value.

4. Payment Behavior

- Analysis of the payments table showed that 100% of transactions recorded at least one installment, indicating that installment-based payments are the universal norm on this platform — a critical insight for financial planning and credit risk models.

5. Monthly Order Trends (2018)

- Order counts were broken down by month for 2018.
- A seasonal pattern was observed, with order volumes peaking mid-year and tapering toward year-end, informing demand forecasting and inventory strategies.

6. Average Products per Order by City

- A CTE-based query computed the average basket size (products per order) at the city level.
- Cities with higher average product counts signal stronger per-session engagement and may represent higher-value customer segments.

7. Price vs. Purchase Frequency Correlation

- A correlation between average product price and order count across categories was computed.
- This analysis helps identify whether higher-priced categories attract fewer buyers (elastic demand) or maintain strong volumes regardless of price (inelastic demand).

8. Seller Revenue Rankings

- Revenue was summed per seller using a window function (DENSE_RANK) to produce a leaderboard of top-performing sellers.
- The top 10 sellers were visualized, highlighting the degree of revenue concentration among high-performing sellers.

9. Cumulative Sales Growth Over Time

- Monthly revenue was tracked cumulatively across years using a running total window function.
- The resulting multi-year line chart clearly illustrates the platform's growth trajectory and highlights inflection points in revenue acceleration.

10. Year-over-Year (YoY) Growth Rate

- Using the LAG window function, annual revenue growth rates were computed between consecutive years.
- This metric quantifies business momentum and is a key indicator for stakeholder reporting and strategic planning.

11. Top Customers by Annual Spend

- The top 5 highest-spending customers were identified per year using DENSE_RANK partitioned by year.
- This high-value customer segmentation supports targeted retention strategies and personalized marketing efforts.

Technical Stack

Tool	Role
Python 3.x	Data orchestration, analysis, visualization
MySQL	Relational data storage and SQL analytics
Pandas	DataFrame manipulation and correlation analysis
Matplotlib	Bar charts, line plots
Seaborn	Styled visualizations with statistical context
mysql-connector-python	Database connectivity

Conclusion

This analysis demonstrates end-to-end proficiency in data engineering, SQL analytics (window functions, multi-table joins), and data visualization. The findings provide a data-driven foundation for decisions across marketing, inventory management, seller partnerships, and customer retention making this a reusable analytical framework adaptable to any e-commerce dataset.